

Lab 1: Introduction to Gridworld and the Agent–Environment Loop

AAI 4323 – Reinforcement Learning

Overview

In this lab, you will implement your first reinforcement learning environment and agent. The goal is to understand the basic structure of the agent–environment loop, how episodes unfold, and how rewards accumulate over time.

You will build and run a simple deterministic Gridworld environment and write a basic agent that interacts with it. This lab sets the foundation for all future labs, including Dynamic Programming, Monte Carlo methods, and Temporal-Difference learning.

Gridworld Environment

You will implement a 4×4 deterministic Gridworld with the following layout:

```
S . . .  
. # . .  
. . . .  
. . T G
```

Legend:

- S – Start state
- . – Empty cell
- # – Wall (impassable)
- T – Trap (terminal, reward = -10)
- G – Goal (terminal, reward = $+10$)

Every non-terminal step yields a reward of -1 . Transitions are deterministic.

Learning Objectives

By the end of this lab, you should be able to:

- Implement a simple Gridworld environment.
- Understand the agent–environment interaction loop.
- Track state transitions, rewards, and episode termination.
- Implement a basic agent (e.g., random agent).
- Run multiple episodes and compute cumulative returns.
- Visualize the agent’s trajectory through the grid.

Files You Will Edit

You will implement your environment and agent in:

```
src/envs/gridworld.py
src/agent.py
```

You will run your code using:

```
src/main.py
```

The autograder tests are located in:

```
tests/test_lab1.py
```

Task 1: Implement the Gridworld Environment

Your environment must support:

- A state representation using (row, col) coordinates.
- A deterministic transition function.
- A reward function:
 - -1 for each step
 - +10 for reaching the goal
 - -10 for stepping on the trap
- Episode termination when reaching T or G.
- A rendering function that prints the grid with the agent marked as A.

Task 2: Implement a Random Agent

Your agent should:

- Select actions uniformly at random from the action set:
Up, Right, Down, Left
- Interact with the environment using the agent–environment loop:

$$s_{t+1}, r_t = \text{env.step}(a_t)$$

- Track cumulative reward for each episode.

Task 3: Run Multiple Episodes

In `main.py`, run at least 10 episodes and print:

- the sequence of states visited
- the sequence of actions taken
- the sequence of rewards
- the total return for each episode

You should observe that random agents often fall into the trap or wander inefficiently.

Task 4: Visualization

Use your environment's rendering function to display the agent's position at each step. A sample rendering might look like:

```
S . . .  
  # . .  
  A . .  
  . . T G
```

Running Your Code

From the project root:

```
python src/main.py
```

This will run your random agent through multiple episodes and print the results.

Grading Rubric

Your work will be graded according to the following rubric:

Component	Points
Environment Implementation	30
Agent Implementation	20
Episode Execution & Logging	25
Visualization	15
Code Quality & Organization	10
Total	100

Submission

Submit:

- `src/envs/gridworld.py`
- `src/agent.py`
- any modifications to `src/main.py`

Do not modify the autograder or the provided starter files.